EVAN T.C. QUINTANA

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SUMMARY

I'm an AI security professional, with a majority of my expertise in technical methods. My recent domains have been a) counter-adversarial ML (CAML) and b) verification for {secure inference, loss-of-control response scenarios, model integrity, multilateral agreements like international compute pacing} and my motivations draw me towards those angles in AI sec which are severely underworked but critical. Some things interesting me right now are: a) downstream impacts on open models and on tech-adoption policies due to developments on the frontier, and b) field-building and systemization for urgently needed technical capabilities (especially cyber-threats).

EXPERIENCE

R&D Data Scientist, AI Security | Sandia National Laboratories, Org. 5563 (Robust Systems Analysis) | *10/26 – current*

- Focuses: CAML, formal methods for verification, complex networks, genAI for national security. Basic research.
- Q+SCI clearance in process.

Graduate R&D Intern | Sandia National Laboratories, Org. 5565 (Math & Analytics for natsec) | *06/26 – 09/26*

- Post-trained an 8B model in a simulated adversarial setting to characterize the possibilities for compromise of safety-tuned base model snapshots. Held role of co-lead on the AI red team within the evaluations group of a division-1400 trilab genAI project.
- Research directions: AI integrity, adversarial robustness, verification of secure compute.

Teaching Assistant, Department of Applied Mathematics | University of California, Santa Cruz | *01/26 – 06/26*

Intern, Global Engagement Center, China Desk (GEC) | U.S. Department of State | *2024*

- Authored 10+ internal reports on malign disinformation campaigns (example country 1: analyzed 600+ national outlets for content-sharing from confirmed global propaganda newswires), distributed to U.S. embassies. Held Secret clearance.

TECHNICAL PROJECTS

Scientific Machine Learning (SciML) | graduate course projects

- Conditional diffusion for sparse-to-full field reconstruction
- Conditional variational autoencoder (CVAE) for autoregressive probabilistic forecasting

Echo-state networks for data-driven superparameterization | undergraduate research

Election disinformation in Taiwan via network analysis | Fulbright semifinalist, open research grant

- Using a 700k-post corpus of social media interactions, I performed community clustering and network analysis to isolate an activity spike around (for example) a false narrative that presidential candidate Lai Ching-Te had an illegitimate child and was unfit for office. RAND research independently identified the same narrative two years later.

AI Red-Teaming Exercise: Catastrophic Risk | Expert participant, Nemesys Insights (under NDA)

EDUCATION

University of California, Santa Cruz: M.S. Scientific Computing and Applied Mathematics, 06/26. GPA 4.0.

University of California, Santa Cruz: B.S. Applied Mathematics, summa cum laude, and B.A. Global Economics. GPA 3.77.

National Taiwan University: Study abroad. Focus: Chinese, political economy of semiconductors.

Graduate coursework: SciML, numerical linear algebra, numerical optimization, applied optimal control, stochastic differential equations, high-performance computing; independent study in uncertainty quantification for SciML.

SKILLS AND CLEARANCE

Technical: Python (pytorch, jax, scipy, xarray, vllm), Rust, Git, Linux.

Languages: English (native), Chinese (proficient), Spanish (proficient).

Clearance: Q+SCI, in progress as of October 26. **Interim Secret**, active from Jan 24 to March 24.

Things that worry me: How do we verify computational infeasibility of a task? What alignment-science paradigm will succeed RL?

TALKS

UCSC Digital Scholarship Symposium: Gave three-minute lightning talk. Presented methods and findings for Taiwan disinfo project.

Koret Research Symposium: Gave a year-end poster session regarding the abovementioned Taiwan disinfo project.